

Blast Furnace Operations & Maintenance Manual

1. Introduction

This manual covers the operations and maintenance of Blast Furnaces BF-001 and BF-002.

2. Normal Operating Parameters

Temperature: 1150-1300 C. Pressure: 150-200 bar. Airflow must be maintained consistently to ensure proper smelting.

3. Tuyere Maintenance

Tuyeres are critical for air injection. Inspect tuyeres weekly for wear and tear. If temperature drops below 1100 C, check for tuyere blockage.

4. Cooling System

The stove cooling system uses water. Check for leaks daily. A leak will cause a sudden pressure drop below 140 bar.

5. Emergency Shutdown

If temperature exceeds 1550 C or pressure exceeds 250 bar, initiate emergency shutdown procedure immediately to prevent explosion.